



# PROBLEM STATEMENT, SCOPE, COMMERCIAL VALUE & SOCIAL IMPACT

## Process Manual Sections 3.1 & 3.2: Rigorous Boundary Definition & Multi-Dimensional Value Proposition

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|----------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Project:</b> Alliance Smart Campus Digital Twin | <b>Faculty Mentor:</b> Dr. Ganga Holi (ASAC CSE)                                                           |
| <b>Team Lead:</b> Likith S (2411021061437)         | <b>Members:</b> Vinodkumar (2411021061425), Nikhil kumar V (2411021061427), Kalyan kumar T (2411021061421) |

### 1. Identification of the Problem Statement (Section 3.1)

In conformance with Section 3.1 of the Process Manual, a clear, relevant, and well-defined problem statement was identified through direct observation of Alliance University's 60-acre Anekal campus infrastructure, interviews with facility engineers, and empirical utility audits. The campus operates high-density academic and residential zones with centralized cooling, a 480 kWp rooftop solar PV array, a 1.2 MWh BESS, and an 11 kV BESCOM grid connection. The fundamental engineering problem is formulated as follows:

**Formal Problem Statement:**  
*"Campus facility directors lack unified, real-time spatial observability to prevent expensive commercial peak-demand electricity penalties and eliminate phantom HVAC energy waste caused by a 28.4% empirical classroom ghost-booking rate. Existing legacy SCADA systems operate in isolated data silos with >10s polling latency and require specialized desktop software, preventing automated, closed-loop solar-battery dispatch and ubiquitous web-accessible governance."*

### 2. Defining Scope, Commercial Value & Social Impact (Section 3.2)

| Project Dimension               | Rigorous Architectural Scope & Boundaries                                                                                                                                                                                                                                                                                                   | Commercial Value & Societal Impact                                                                                                                                                     |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope & Boundaries              | <ul style="list-style-type: none"><li>In-Scope: Procedural Blender 3D modeling of 6 core campus zones; WebGL 60 FPS rendering; 1,024-node simulated IoT telemetry; BESS peak shaving; anonymous vacancy tracking.</li><li>Out-of-Scope: 11 kV physical breaker tripping; off-campus municipal infrastructure; facial recognition.</li></ul> | Eliminates scope creep by establishing measurable boundaries; ensures complete execution within one academic semester.                                                                 |
| Commercial Value                | Coordinated solar-BESS peak shaving reduces peak commercial grid draw by 18.4% (from 684 kW to 558 kW). Automated ghost-booking reclamation cuts unoccupied lecture hall chiller load by 22.6%.                                                                                                                                             | Generates projected electricity bill savings of INR 18,400/day under BESCOM commercial tariffs (~INR 55.2 Lakhs annually). Reclaims 4.2 hours of classroom capacity per room per week. |
| Societal & Environmental Impact | Decarbonizes higher-education campus operations by optimizing renewable solar self-consumption from 74.2% to 91.8%. Guarantees privacy through strictly anonymous headcount sensing.                                                                                                                                                        | Reduces institutional carbon emissions by 142 tons of CO <sub>2</sub> equivalent annually; establishes a replicable open-source smart campus framework for Indian universities.        |